DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
SCIENCE DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-CSPE004
COURSE TITLE	**:**	CS Professional Elective 3
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	CS Professional Elective 1, 2
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Rolando Barrameda
CONSULTATION HOURS	**:**	Monday: (2:00 – 4:00 PCH101) | Tuesday: (2:00 – 4:00 PCH101)
CONTACT INFORMATON	**:**	rbbarrameda@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides an introduction to the field of Artificial Intelligence (AI), focusing on its application to solve real-world problems that are challenging to address using traditional algorithmic approaches. Students will explore key AI methodologies, including techniques for handling uncertainty, learning from experience, and applying nature-inspired problem-solving strategies. The course covers both the theoretical foundations and practical aspects of developing intelligent systems. Additionally, it addresses the ethical considerations in AI development, preparing students for careers in AI, data science, and related fields.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. acknowledge the foundational principles of artificial intelligence in a broad context, including exploration of selected real-world applications;

CLO 2. demonstrate the various techniques and methods that algorithms in artificial intelligence employ within the context of Data Science;

CLO 3. utilize expertise in intelligent systems, encompassing AI, Machine Learning, and Deep Learning;

CLO 4. develop an application program for a designated intelligent system project using programming language tools, and subsequently compose a research paper and deliver a presentation; and

CLO 5. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. acknowledge the foundational principles of artificial intelligence in a broad context, including exploration of selected real-world applications.	TLO1. Define AI and explain how it can apply in many fields of work. TLO2. Describe the different applications and their functionalities within the realm of artificial intelligence.
CLO2. demonstrate the various techniques and methods that algorithms in artificial intelligence employ within the context of Data Science.	TLO3. Identify the different categories of applications for Intelligent AI and Data Science. TLO4. Perform a comparative analysis and research on the different methodologies and techniques utilized in intelligence systems.
CLO 3. utilize expertise in intelligent systems, encompassing AI, Machine Learning, and Deep Learning.	TLO5. Demonstrate how various Artificial Intelligence algorithm work using the Spider, Collab, MATLAB and other related software.

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 0 TLO1	Week 1 Jan 20 – 25	• Gospel Reading, Reflection, and Prayer • Presentation of course introduction • Module 1 • Introduction to Data Science	Power point MS Team Spider/Colab	✓		2

Module 1 TLO1 TLO2 TLO3 TLO4 TLO5 CLO1 CLO2 CLO3	Week 2 Jan 25 – Feb 2	• Gospel Reading, Reflection, and Prayer • Module 1 • Artificial Intelligence Intuition Enabling assessment: Explain and Identify the rules of CNN (15 pts.) Short quiz on Data Science (20 pts.)	Power point MS Team Spider/Colab	✓		2
Module 1 TLO1 TLO2 TLO3 TLO4 CLO1 CLO2 CLO3	Week 3 Feb. 3-8	• Gospel Reading, Reflection, and Prayer • Module 1: Deep Learning Enabling assessment: Discuss the role of Deep Learning in other domain (20 pts.)	Power point MS Team Spider/Colab		✓	2
Module 2 TLO1 TLO2 TLO3 TLO4 TLO5 CLO1 CLO2 CLO3	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Module 2: Fuzzy Logic Technique Long Quiz on module 1 and 2 (45 pts)	Power point MS Team Spider/Colab	✓		2
Module 2 TLO1 TLO2 TLO3 TLO4 TLO5 CLO1 CLO2 CLO3	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • Module 2: Continuation of Discussion in Fuzzy Logic	Power point MS Team Spider/Colab		✓	2

	Week 6 Feb 24 – March 1	• Gospel Reading, Reflection, and Prayer • Module 3: Data Wrangling and Analysis Enabling assessment: Connect and analyze the data Wrangling (15 pts.)	Power point MS Team Spider/Colab	✓		2
Module 3 TLO1 TLO2 TLO3 TLO4 TLO5	Week 7 March 3-8	• Gospel Reading, Reflection, and Prayer Utilization of DLSUD Library			✓	2
	Week 8 March 10-14	• Gospel Reading, Reflection, and Prayer Research Collab, Preparation for the Institutional Research Week (3rd INCRIS Conference)	Synchronous Class		✓	2
CLO1 CLO2 CLO3	Week 9 March 15 Oct 17-21	Midterm Summative Exam covers Module 1, 2, 3		✓		2
TOTAL				10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 3. utilize expertise in intelligent systems, encompassing AI, Machine Learning, and Deep Learning.	TLO5. Demonstrate how various Artificial Intelligence algorithm work using the Spider, Collab, MATLAB and other related software.
CLO 4. develop an application program for a designated intelligent system project using programming language tools, and subsequently compose a research paper and deliver a presentation.	TLO6. Develop a functional application program to be included in the research project and presentation in local and international publication.

CLO 5. demonstrate 21st century skills in all learning activities.	TLO7. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 4 TLO5 TLO6 TLO7 CLO3 CLO4 CLO5	Week 10 March 24-29	Sustainability Week	Power point MS Team Power BI	✓		2
Module 4 TLO5 TLO6 TLO7 CLO3 CLO4 CLO5	Week 11 March 31 – April 5	Module 4: Data Visualization and Power BI Enabling assessment: Classify and recognize power bi assets Short quiz on Power BI (15 pts)	Power point MS Team Power BI		✓	2
Module 5 TLO5 TLO6 TLO7	Week 12 April 7 -11	Gospel Reading, Reflection, and Prayer Module 5: Data Analytics Short quiz on Analytics (15 pts.)	Power point MS Team Spider/Colab	✓		2

CLO3 CLO4 CLO5						
Module 6 TLO3 TLO4	Week 13 April 14 - 19	• Gospel Reading, Reflection, and Prayer • Independent Learning Week	Power point MS Team Spider/Colab	✓		2
TLO5 TLO6 TLO7	Week 14 April 21-25	• Gospel Reading, Reflection, and Prayer • Research Week • Research Paper Presentation 3rd Incris Conference	Power point MS Team Spider/Colab	✓		2
CLO3 CLO4 CLO5	Week 15 April 28 – May 3	Gospel Reading, Reflection, and Prayer Module 6: Research Research Publication (Ject and other reputable publications)	Power point MS Team Spider/Colab		✓	2
	Week 16 May 5 - 10	Final Summative Examination on Module 4, 5 Graduating Students	Power point MS Team Spider/Colab	✓		2
	Week 17 Dec 12-17	• Gospel Reading, Reflection, and Prayer • Completion Week	Power point MS Team Spider/Colab			2

	Week 18 Dec 16-18	• Gospel Reading, Reflection, and Prayer • Completion Week				2
TOTAL				10	8	18

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Library	Hajian A, Baloian N, Inoue T, Luther W. 2022. Data science, human-centered computing, and intelligent technologies. Berlin: Logos Verlag.
AEA Library	Rivas P. 2020. Deep learning for beginners: A beginner's guide to getting up and running with deep learning from scratch using Python. Packt Publishing. [accessed 2024 Jan 22].
AEA Library	Randolph T. 2021. Artificial intelligence: Work, machines and human interaction. New York: SNOVA. [accessed 2024 Jan 22].
e-book 3541254	Hajian A, Baloian N, Inoue T, Luther W. 2022. Data science, human-centered computing, and intelligent technologies [Internet]. Berlin: Logos Verlag. [cited 2024 Jan 22]. Available from: https://eds.p.ebscohost.com/eds/detail/detail?vid=1&sid=f2ccd5ba-1f03-43d2-a3b6-

	34daff0dffda%40redis&bdata=JkF1dGhUeXBlPXNoaWImc2l0ZT1lZHMtbGl2ZQ%3d%3d#AN=3541254&db=nlebk

Online References

Call Number/ E-provider	Reference Material
Online	Simplilearn. 2023. Basics of data analytics. Data analytics fundamentals and data analytics training [Internet]. [accessed 2024 Jan 22]. Available from: https://www.simplilearn.com/tutorials/machinelearning-tutorial/what-is-machinelearning
YouTube	IU Luddy. [date of publication]. Intelligent systems research [Internet]. Bloomington (IN): Indiana University Luddy School of Informatics, Computing, and Engineering. Available from: https://www.youtube.com/watch?v=oHsVAtUUodU

Prepared: **Computer Science Department**
2nd Semester 2024-2025

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